

Comprehensive Cardiomyopathy Panel

Patient name: XX	PIN: XX
Age: 9 Months	Sex: Male
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: XX
Specimen: XX	Sample Received Date: XX
	Report Date: XX

Clinical History:

Proband, XX was first born to non-consanguineous parents. He was preterm baby delivered via LSCS, birth weight was 2.26 kgs and cried immediately at birth. He presented with dilated cardiomyopathy, congestive heart failure, and bronchiolitis. His ECHO was indicative of global hypokinesis of dilated LV, severe LV dysfunction, pHTN EF - 18 %. X Ray indicative of cardiomegaly and his repeat ECHO was indicative of situs solitus, levo cardia, globular hypokinesia, EF - 25 %, Grade 2 LVD - s/o Idiopathic dilated cardiomyopathy. His mother had gestational diabetes mellitus. His TORCH profile, HBV, HCV IgG was negative and PCR for Adenovirus, Parvo virus, EBV, HHV-6 was negative. Proband, XX is suspected to have dilated cardiomyopathy and have been evaluated for pathogenic variations.

Results

Likely pathogenic variant related to given phenotype was identified

List of significant variant identified related to the phenotype:

Gene	Variant*	Allele Status	Region	Disease	Inheritance	Classification*
MYH7 (-) (NM_000257.4)	c.1618T>C (p.Phe540Leu)	Heterozygous	Exon 16	Congenital myopathy 7A, myosin storage, autosomal dominant (OMIM# 608358) Cardiomyopathy, dilated, 1S (OMIM# 613426) Cardiomyopathy, hypertrophic, 1 (OMIM# 192600) Laing distal myopathy (OMIM# 160500)	Autosomal Dominant	Likely Pathogenic

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
 No other variant that warrants to be reported for the given clinical indication was identified.

Variant Interpretation:

Gene	Variant*	Inheritance	Disorder or Phenotype	Criteria	Classification*
MYH7 (-) (NM_000257.4)	c.1618T>C (p.Phe540Leu)	Autosomal Dominant	Congenital myopathy 7A, myosin storage, autosomal dominant (OMIM# 608358) Cardiomyopathy, dilated, 1S 608358 (OMIM# 613426) Cardiomyopathy, hypertrophic, 1 (OMIM# 192600) Laing distal myopathy (OMIM# 160500)	PM1, PM2, PP3, PP5	Likely Pathogenic
Region	Allele State	Allelic Read Depths			
Exon 16	Heterozygous	Ref(A): 73, Alt(G): 69, VAF: 48.59%			
Genomic Position		Variant Frequency			
Chr14:NC_000014.9:g.23427855A>G		Not identified in large population studies			

Variant summary: A heterozygous missense variant in exon 16 of MYH7 gene (Chr14:NC_000014.9:g.23427855A>G; NM_000257.4; Depth: 142x) that results in amino acid substitution of Leucine for Phenylalanine at codon 540 (p.Phe540Leu) was detected.

Clinical and Literature evidence: This variant has been previously classified as likely pathogenic variant in ClinVar database (Variation id: [407186](#)). This variant has been reported in patient with dilated cardiomyopathy (PMID: [26899768](#)).

Population Frequency: This variant has been not observed in the large population cohorts of gnomAD v4 All, or 1kG All.

In-silico prediction: The *in-silico* predictions of the variant are damaging by SIFT and PolyPhen2 and deleterious by Mutation Taster2. The reference codon is conserved across mammals in PhyloP and GERP++ tools.

OMIM phenotype: Congenital myopathy 7A, myosin storage, autosomal dominant (OMIM#[608358](#)). Cardiomyopathy, dilated, 1S (OMIM#[613426](#)), Cardiomyopathy, hypertrophic, 1 (OMIM#[192600](#)) and Laing distal myopathy (OMIM#[160500](#)) are caused by heterozygous mutation in the MYH7 gene (OMIM#[160760](#)). Autosomal dominant myosin storage congenital myopathy-7A (CMYO7A) is a skeletal muscle disorder with wide phenotypic variability. The age at symptom onset can range from early childhood to late adulthood. Affected individuals have proximal muscle weakness affecting the upper and lower limbs and distal muscle weakness of the lower limbs, resulting in gait difficulties and scapular winging (scapuloperoneal myopathy). Additional features may include thin habitus, high-arched palate, foot drop, pes cavus, calf pseudohypertrophy, and decreased reflexes. The severity is also variable: some patients develop respiratory insufficiency, joint contractures, and scoliosis in the first decades, whereas others are clinically unaffected, but show subtle signs of the disorder on examination. Serum creatine kinase may be normal or elevated. The disease is usually slowly progressive and most patients remain ambulatory. Skeletal muscle biopsy can show different abnormalities, including hyaline bodies, type 1 fiber predominance, congenital fiber-type disproportion (CFTD), and nonspecific myopathic changes with myofibrillar disarray. These diseases follow autosomal dominant pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as likely pathogenic variant. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of the finding. If the results do not correlate, additional testing may be considered based on the phenotype observed.**

Methodology:

DNA extracted from the blood was used to perform whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80 to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAScope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In silico predictions of the variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWMSplice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes:

Total Read Generated	16.04 Gb
Data ≥ Q30	96.96%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer:

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.

- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

References:

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Miller, David T., et al. "ACMG SF v3. 1 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in Medicine 24.7 (2022): 1407-1414.

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Appendix 1: Gene Coverage

Indication Base Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
A2ML1	100.0	ABCC9	100.0	ABCD1	100.0	ACAD9	100.0
ACADVL	100.0	ACSL4	100.0	ACTA2	100.0	ACTB	100.0
ACTC1	100.0	ACTG1	100.0	ACTN2	100.0	ACVR2B	100.0
ACVRL1	100.0	ADA2	100.0	AFF2	100.0	AGL	100.0
AGTR2	100.0	AKAP9	100.0	AKT3	100.0	ALG12	100.0
ALG3	100.0	ALG6	100.0	ALMS1	100.0	ANK2	100.0
ANK3	100.0	ANKRD1	100.0	ANKRD17	100.0	AP1S2	100.0
AP4B1	100.0	AP4E1	100.0	AP4M1	100.0	AP4S1	100.0
APOA5	100.0	APOB	100.0	ARHGEF6	100.0	ARID2	100.0
ARL6	100.0	ASXL1	100.0	ASXL2	100.0	ATP13A2	100.0
B4GALT7	100.0	B9D1	100.0	B9D2	100.0	BAG3	100.0
BBS1	100.0	BBS10	100.0	BBS12	100.0	BBS2	100.0
BBS4	100.0	BBS5	100.0	BBS7	100.0	BBS9	100.0
BCAP31	100.0	BCL11A	100.0	BCOR	100.0	BMPR2	100.0
BPTF	99.64	BRAF	100.0	BRD4	100.0	BRSK2	99.85
BRWD3	100.0	C1R	100.0	C1S	100.0	CACNA1C	100.0
CACNA2D1	97.73	CACNB2	100.0	CACNG2	100.0	CALM1	100.0
CALM2	100.0	CALM3	100.0	CALR3	100.0	CASQ2	100.0
CAV1	100.0	CAV3	100.0	CAVIN4	100.0	CBL	100.0
CC2D1A	100.0	CC2D2A	100.0	CCDC103	100.0	CCDC22	100.0
CCDC28B	100.0	CCDC39	100.0	CCDC40	100.0	CCDC65	100.0
CCN1	100.0	CCNO	100.0	CDH15	100.0	CEP290	100.0
CFAP53	100.0	CHD3	100.0	CHD7	100.0	CHRM2	100.0
CITED2	100.0	CLIC2	100.0	CNKSR2	100.0	CNOT1	100.0
CNOT3	100.0	CNTN6	100.0	COL3A1	100.0	COL5A1	100.0
COL5A2	100.0	COX15	100.0	CPS1	100.0	CPT1A	100.0
CPT2	100.0	CRADD	100.0	CRBN	100.0	CRELD1	100.0
CRYAB	100.0	CSNK2A1	100.0	CSRP3	100.0	CTF1	99.78
CTNNA3	100.0	CUL3	100.0	CUL4B	100.0	DEPDC5	100.0
DES	100.0	DKC1	100.0	DLG4	100.0	DLGAP2	100.0
DMD	100.0	DNAAF1	100.0	DNAAF2	100.0	DNAAF3	100.0
DNAAF4	100.0	DNAAF5	100.0	DNAH11	100.0	DNAH5	100.0
DNAH8	100.0	DNAI1	100.0	DNAI2	100.0	DNAJC19	100.0
DNAL1	100.0	DOLK	100.0	DPP6	99.77	DRC1	100.0
DSC2	100.0	DSG2	100.0	DSP	100.0	DTNA	100.0
EBP	100.0	EED	100.0	EFEMP2	100.0	EIF2S3	100.0
ELAC2	100.0	ELN	100.0	EMD	100.0	EN2	100.0
ENG	100.0	EPB41L1	100.0	EXOC7	100.0	EYA4	100.0
EZH2	100.0	FBN1	100.0	FBN2	100.0	FGD1	100.0
FHL1	100.0	FHL2	100.0	FKRP	100.0	FKTN	100.0
FLNA	100.0	FLNC	100.0	FOXF1	100.0	FOXH1	100.0
FOXP2	100.0	FRMPD4	100.0	FTSJ1	100.0	FXN	100.0
GAA	100.0	GATA4	100.0	GATA6	100.0	GATAD1	100.0
GDI1	100.0	GJA1	100.0	GJA5	100.0	GK	100.0
GNAI1	100.0	GPC3	100.0	GPD1L	100.0	GRIA3	100.0
GRIK2	100.0	GYG1	100.0	H1-4	100.0	HAMP	100.0
HAND1	100.0	HCFC1	100.0	HCN4	100.0	HECW2	100.0
HFE	100.0	HJV	100.0	HNRNPK	100.0	HNRNPR	100.0
HOXA1	100.0	HPRT1	100.0	HRAS	100.0	IDS	100.0
IGBP1	100.0	IL1RAPL1	100.0	ILK	100.0	INVS	100.0
JAG1	100.0	JPH2	100.0	JUP	100.0	KAT6B	100.0
KATNAL2	95.02	KCNA5	100.0	KCND3	100.0	KCNE2	100.0
KCNE3	100.0	KCNE5	100.0	KCNH1	100.0	KCNH2	100.0
KCNJ2	100.0	KCNJ5	100.0	KCNJ8	100.0	KCNK3	100.0
KCNQ1	100.0	KCNQ2	100.0	KCNQ3	100.0	KCNT1	100.0
KDM5B	100.0	KDM6B	100.0	KIF7	100.0	KMT2B	99.51

KMT2C	100.0	KMT2E	100.0	KRAS	100.0	L1CAM	100.0
LAMA4	100.0	LAMC3	100.0	LAMP2	100.0	LAS1L	100.0
LDB3	100.0	LDLR	100.0	LDLRAP1	100.0	LEFTY2	100.0
LINS1	100.0	LMNA	100.0	LZTR1	100.0	MAGEL2	100.0
MAGT1	100.0	MAN1B1	100.0	MAOA	100.0	MAP2K1	100.0
MAP2K2	100.0	MBTPS2	100.0	MCIDAS	100.0	MCPH1	100.0
MED12	100.0	MED13	100.0	MED13L	100.0	MED23	100.0
MEIS2	100.0	MET	100.0	MFAP5	100.0	MIB1	100.0
MID1	100.0	MKKS	100.0	MKS1	100.0	MMP21	100.0
MRPL3	100.0	MSL3	100.0	MTO1	100.0	MYBPC3	100.0
MYH11	100.0	MYH6	100.0	MYH7	100.0	MYL2	100.0
MYL3	100.0	MYLK	100.0	MYLK2	100.0	MYOM1	100.0
MYOZ2	100.0	MYPN	100.0	NAA15	100.0	NBEA	100.0
NDP	100.0	NDUFAF1	100.0	NDUFB11	100.0	NDUFS2	100.0
NEBL	100.0	NEK8	100.0	NEXMIF	100.0	NEXN	100.0
NF1	100.0	NHS	100.0	NIPBL	100.0	NKX2-5	100.0
NKX2-6	100.0	NLGN3	100.0	NLGN4X	100.0	NME8	100.0
NODAL	100.0	NONO	100.0	NOTCH1	100.0	NPHP3	100.0
NPPA	100.0	NR2F2	100.0	NR4A2	100.0	NRAS	100.0
NSD1	100.0	NSD2	100.0	NSDHL	100.0	NSUN2	100.0
NTRK3	100.0	OCRL	100.0	OFD1	100.0	PAK3	100.0
PCDH19	100.0	PCSK9	100.0	PDLIM3	100.0	PGAP3	100.0
PHF21A	100.0	PHF6	100.0	PHF8	100.0	PHIP	100.0
PIGL	100.0	PIGN	100.0	PIK3CA	100.0	PIK3R2	100.0
PITX2	100.0	PKD1L1	100.0	PKP2	100.0	PLOD1	100.0
POLG	100.0	PORCN	100.0	PPM1D	100.0	PPP2R1A	100.0
PQBP1	100.0	PRDM16	100.0	PRKAG2	100.0	PRKG1	100.0
PRPS1	100.0	PRRT2	100.0	PRSS12	100.0	PSEN2	100.0
PTCHD1	100.0	PTPN11	100.0	PUF60	100.0	QRICH1	100.0
RAB39B	100.0	RAB40AL	100.0	RAD21	100.0	RAF1	100.0
RAI1	100.0	RASA1	100.0	RBM10	100.0	RBM20	100.0
REER	100.0	RIT1	100.0	RPGRIP1L	100.0	RPL10	100.0
RRAS	100.0	RYSR2	100.0	SARS1	100.0	SCN10A	100.0
SCN1A	100.0	SCN1B	100.0	SCN2B	100.0	SCN3B	100.0
SCN4B	100.0	SCN5A	100.0	SCN8A	100.0	SCN9A	100.0
SDHA	100.0	SEMA3E	100.0	SET	100.0	SETBP1	100.0
SGCD	100.0	SHANK2	100.0	SHOC2	100.0	SHROOM4	100.0
SIN3A	100.0	SKI	100.0	SLC22A5	100.0	SLC25A20	100.0
SLC2A1	100.0	SLC2A10	100.0	SLC40A1	100.0	SLC6A4	100.0
SLC9A9	100.0	SLMAP	100.0	SMAD3	100.0	SMAD4	100.0
SMAD6	100.0	SMAD9	100.0	SMARCA1	100.0	SMARCC2	100.0
SMC3	100.0	SNAP25	100.0	SNTA1	100.0	SOBP	99.71
SON	100.0	SOS1	100.0	SOS2	100.0	SOX5	100.0
SPAG1	100.0	SPRED1	100.0	STAG1	100.0	SYNE1	100.0
SYP	100.0	TAB2	100.0	TANC2	100.0	TAOK1	100.0
TBR1	100.0	TBX1	98.22	TBX20	100.0	TBX5	100.0
TCAP	100.0	TCF7L2	100.0	TCTN2	100.0	TECR	100.0
TELO2	100.0	TFR2	100.0	TGFB2	100.0	TGFB3	100.0
TGFBR1	100.0	TGFBR2	100.0	TLL1	100.0	TMEM216	100.0
TMEM231	100.0	TMEM43	100.0	TMEM67	100.0	TMEM70	100.0
TMPO	100.0	TNNC1	100.0	TNNI3	100.0	TNN2	100.0
TNRC6B	100.0	TPM1	100.0	TRAPPC9	100.0	TRDN	95.24
TRIM8	100.0	TRIP12	100.0	TRPM4	100.0	TRRAP	100.0
TSPAN7	100.0	TTC8	100.0	TTN	99.9	TTR	100.0
TUSC3	100.0	TXNRD2	100.0	UPF3B	100.0	USP27X	100.0
USP7	100.0	VCL	100.0	WAC	100.0	WDFY3	100.0
WDPCP	100.0	YWHA	100.0	ZC3H14	100.0	ZC4H2	100.0
ZDHHC15	100.0	ZDHHC9	100.0	ZFPM2	100.0	ZIC3	100.0
ZMIZ1	100.0	ZMYND10	100.0	ZNF292	100.0	ZNF41	100.0
ZNF674	100.0	ZNF711	100.0	ZNF81	100.0	MT-TF	100.0
MT-RNR1	100.0	MT-TV	100.0	MT-RNR2	100.0	MT-TL1	100.0
MT-ND1	100.0	MT-TI	100.0	MT-TQ	100.0	MT-TM	100.0
MT-ND2	100.0	MT-TW	100.0	MT-TA	100.0	MT-TN	100.0
MT-TC	100.0	MT-TY	100.0	MT-CO1	100.0	MT-TS1	100.0
MT-TD	100.0	MT-CO2	100.0	MT-TK	100.0	MT-ATP8	100.0

MT-ATP6	100.0	MT-CO3	100.0	MT-TG	100.0	MT-ND3	100.0
MT-TR	100.0	MT-ND4L	100.0	MT-ND4	100.0	MT-TH	100.0
MT-TS2	100.0	MT-TL2	100.0	MT-ND5	100.0	MT-ND6	100.0
MT-TE	100.0	MT-CYB	100.0	MT-TT	100.0	MT-TP	100.0